Linux System Administration (LSA)

Course Introduction

SoftUni Team

Technical Trainers

Software University

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https://softuni.bg

Have a Question?

Software University

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#LSA

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Course Objectives and Scope

What and When Will Be Covered

Course Objectives

- Installation and Configuration of Linux Servers
- Feel Comfortable to Work in the Console
- Users and Access Management
- Software and Services Management
- Network Management
- BASH Scripting and Basic Tasks Automation

Course Scope and Schedule

- **Week #1**

Module 1

- Introduction to Linux
- Virtualization and Installation of a Linux distribution
- First Steps in the Console

Module 2

- Console Navigation. Getting Help
- Working with Files and Folders
- Access Rights. Users and Groups

Course Scope and Schedule

- **Week #2**

Module 3

- Text Editors. Regular Expressions
- Search for Files. Extract Information from Files
- Standard Streams. Command Chains

Module 4

- Network and Network Services Management
- Packages. Package Managers. Software Management
- Services Management

Course Scope and Schedule

- **Week #3**

Module 5

- Boot Process. Boot Loaders and Managers
- System Startup Process. Runlevels / Targets
- Processes and Resources Monitoring

Module 6

- Disks and Partition Management
- File Systems and FHS
- Backup and Restore Files

Course Scope and Schedule

- **Week #4**

Module 7

- BASH in Detail
- Script Creation
- Tasks Automation

Module 8

- Course Overview
- Exam Preparation

We Will Cover

- More than 250 commands
- 3 distribution families
- 3 file systems
- … and many other interesting things

Is This Enough? *

- The short answer is **NO** (or **IT DEPENDS**) ☺
- **Learning by doing** is the key
- Experiment with **different distributions** **
- Use **various platforms** (virtual and physical)
- Try **different scenarios** and solve a **variety of tasks**
- Read a **book** or **two** (or **even more**) on the subject
- Attend **other courses** on the subject as well

* Soon a Linux Essentials study group will be organized. Stay tuned ;)

** After you feel comfortable enough with one distribution

Requirements

Required Resources

- CPU with VT-x/AMD-V
- 4 GB+ of Free RAM
- 80 GB+ of Free HDD Space
- Virtualization Software (**VirtualBox**, VMware …, Hyper-V)
- ISOs for AlmaLinux OS*, Debian*, and openSUSE*

* **Install Media:** ~ 100 MB - 8 GB/ISO
* **Virtual Machines :**~ 4 GB+/VM
* **Templates:** ~ 4 GB+/VM
* **VM Snapshots:** it depends

*** You can experiment with others as well, but mind that there could be some differences**

Required* Knowledge

- At least intermediate computer literacy
- Basic operating systems knowledge
- Basic networking knowledge

* It is nice to have

Additional Details

Linux System Administration Course – Timeline

17-Mar-2026

19-Apr-2026

26-Apr-2026

Linux System Administration

5 weeks * 2 times / week

6 credits

Start: 17-March-2026

Regular Exam: 19-April-2026
Retake Exam: 26-April-2026

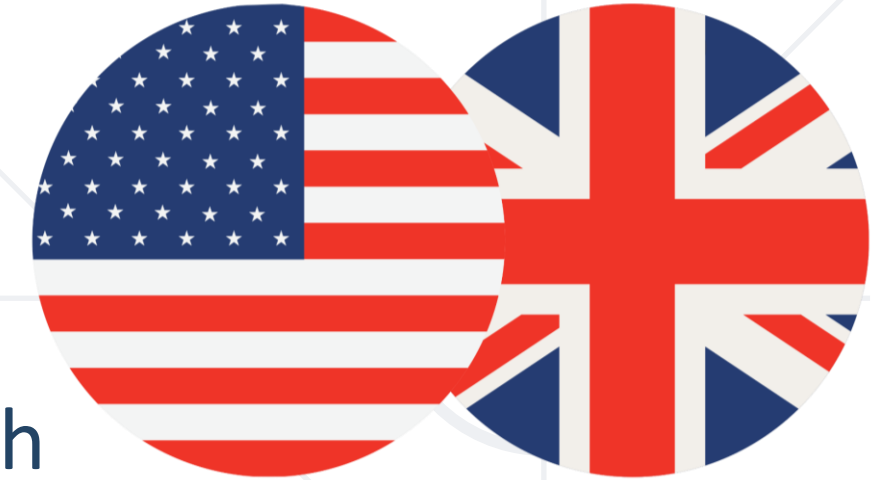
Training Duration

- Lectures: **~ 12 hours** (online)
- Practical exercises (online): **~ 12 hours**
- Self-Exercise: **~ 12 hours**
- Schedule: **March 2026 – April 2026**
  - Every **Tuesday** and **Thursday** between **18:30** and **22:30**
- Exam: **19 April 2026**
- Retake exam: **26 April 2026**

18:30 – 19:00 Part 1
19:00 – 19:15 Break 1
19:15 – 20:15 Part 2
20:15 – 20:30 Break 2
20:30 – 22:30 Part 3

Why English?

- Why the slides are in English?
  - English is the native language of the IT people
  - Specific terminology should be in English
    - Translations are inaccurate and funny
- Just learn English!
  - No excuses



Learn to Search on the Internet

- The course assignments require to **search on the Internet**
  - This is an important part of the learning process
  - Some exercises intentionally have no hints

- Learn to find solutions!
  - Working in IT requires **everyday searching and learning**
  - No excuses, just **learn to study**!
  - IT people learn new technologies and tools every day!

Course Web Site and FB Group

- Official web site:

https://softuni.bg/trainings/5028/linux-system-administration-march-2026

- Official Facebook group:

https://www.facebook.com/groups/linuxsystemadministrationmarch2026

Exam and Evaluation Criteria

Scoring for Linux System Administration

Exam: Practice*

- Manage and configure a set of machines
- Install and configure **a set of software** and services
- Create a set of **files, folders, users and groups**
- Set the **appropriate permissions**
- **Write a script** that does what requested
- Additional tasks **as per the exam requirements**

The **practice exam** will be held **remotely** in a **controlled environment**

All you need is just a PC with **SSH client** and **Internet connectivity**

* Check the **Exam Blueprint** document in the **Course Introduction** section available at:
https://softuni.bg/trainings/5028/linux-system-administration-march-2026#lesson-104772

Exam: Test

20

minutes

20

single-choice questions

Certifications

Introduction to Certifications

- **Authorities**
  - LPI, CompTIA, Linux Foundation, Red Hat, SUSE, Oracle, …
- Categories
  - **By Validity** – Lifetime, Fixed Term, or Fixed Version
  - **By Distribution** – Agnostic or Not
  - **By Type** – Question (test) based, or Performance based
  - **By Place** – At home or at an Exam center
  - **By number of exams**

Linux Professional Institute (LPI) *

- **LPI Linux Essentials**
  - Entry level lifetime certificate
  - Distribution agnostic, question based, 1 exam (010-160)
- **LPIC-1: Linux Administrator**
  - Associate level certificate with 5 years validity
  - Distribution agnostic, question based, 2 exams (101/102-500)
- **LPIC-2: Linux Engineer** – 5 years, 2 exams (201/202-450)
- **LPIC-3: Linux Enterprise Professional** – 5 years, 4 types, 1 ex/t

http://www.lpi.org/our-certifications/summary-of-certifications

Other Certificate Authorities

- **CompTIA** (Linux+)
  - https://www.comptia.org/en-us/certifications/linux/
- **Linux Foundation**
  - https://training.linuxfoundation.org/certification-catalog
- **Red Hat**
  - https://www.redhat.com/en/services/certifications
- **SUSE**
  - https://www.suse.com/training/exam/
- **Oracle**
  - https://education.oracle.com/oracle-certification-path/pFamily_358

The Team

Dimitar Zahariev

- During office hours work as **Consultant** and **Trainer**
- Passionate about technology and teaching
- Both **Microsoft Certified Trainer** and **LPI Approved Trainer**
- Other side activities:
  - **openSUSE Advocate**
  - **VMUG Bulgaria Co-Leader**

About You

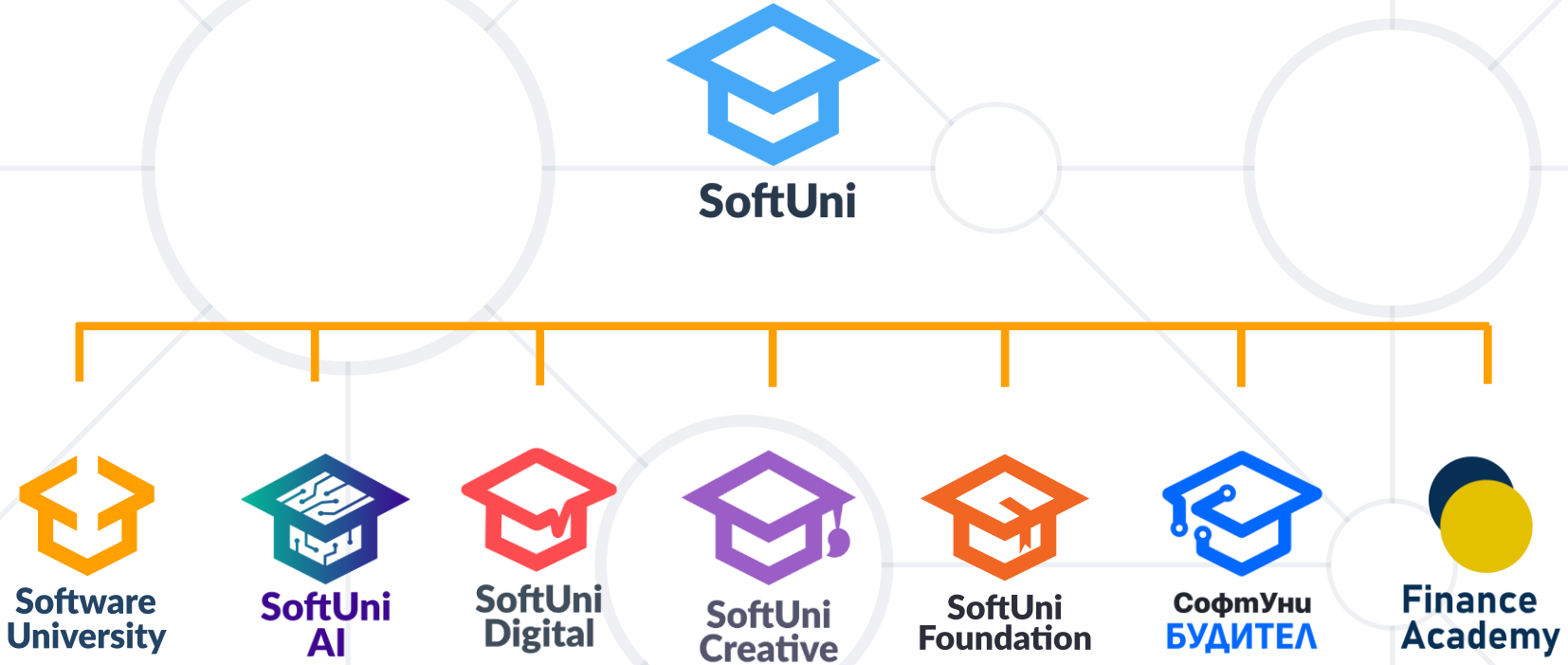
About You

- Go here

  https://zahariev.pro/go/lsa-form

- And share a few words about your
  - Environment
  - Experience
  - Expectations

Questions?



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